Introduction to CSS Projekt

→ general paper for use of google trend data
https://pmc.ncbi.nlm.nih.gov/articles/PMC6246971/?utm_source=chatgpt.com

Independent Variable:
Exposure to a cybersecurity/privacy incident (`PostEvent`, event date, severity).

Dependent Variables:
SVI-levels, peak magnitude, decay time, substitution index, group differences (privacy apps vs controls).

Hypotheses:

1. Incidents increase search interest for privacy-related terms.
2. Privacy apps rise more than mainstream apps.
3. Severity predicts peak height.
4. Interest decays within weeks.
5. Substitution effects between privacy apps and mainstream apps.

Prediction:
Strong but short-lived spikes; privacy apps benefit; control terms remain stable.

Use Google Trends as a proxy for public attention to the Cambridge Analytica scandal and then test whether that attention leaves a measurable imprint on downstream outcomes (e.g., political-advertising spend, stock-market reactions, or voter-behavior surveys).

Typical workflow

Select keywords – “Cambridge Analytica”, “Facebook data scandal”, “micro-targeting”. Choose the same terms across all regions you will compare. 1
Download weekly relative-interest data (0–100) for the full period you wish to study (e.g., 2016 – 2024). 1
Normalize outcome variables (advertising-spend, election-result margins, tech-stock returns) to the same 0–100 scale via min-max scaling. 1
Identify the surge and decay – locate the peak (late 2018) and set a post-surge threshold (e.g., < 20 % of peak) to define the “after-effect” window. 1
Benchmark methodology – apply one of the following published approaches:

Dynamic-interdependency analysis – the ACM-TMIS paper “Election Interference and Online Propaganda Campaigns” builds a VAR/Granger-causality model linking Facebook activity, Google-Trends interest, and New-York-Times coverage to assess cross-platform influence. 2 3

Cluster-based event study – Jeleskovic & Wan (2024) examine the stock-market impact of the scandal with a multi-cluster event-study that isolates abnormal returns around the data-leak dates. 4 5
Comparative microtargeting assessment – Jakee & Fink (2024) compare Cambridge Analytica’s processes to other firms using a qualitative-explanatory framework and then quantify post-event advertising spend. 6
Google-Trends election-forecasting template – the Big-Data journal review outlines how to fit ARIMA or Bayesian structural-time-series models to trends and test lagged correlations with election outcomes. 1

Statistical testing – run cross-correlation or lagged regression between the post-surge trend and each outcome; significance indicates a lasting impact.

Interpretation – if the trend stabilises at a non-zero level and retains significant correlation ($p < 0.05$) with the outcome, the scandal’s influence persists; otherwise the attention has faded.

1. Pick a single, well-defined query (e.g., “Cambridge Analytica”) and download the weekly Google Trends index for the full period you need (0–100 scale).
2. Use only publicly available series that are already archived:
   * Election results by district (official election-commission sites or Wikipedia tables).
   * Stock-price histories for Facebook/Meta (Yahoo Finance CSV).

   → https://www.kaggle.com/datasets/adilshamim8/meta-stock-price-history

   * Advertising-spend summaries from industry reports (e.g., Interactive Advertising Bureau PDFs).
   * Media-coverage counts from Google News Archive (simple article-frequency counts).

3. Normalize each series to 0–100 with min-max scaling so they are comparable with the Trends index.
4. Detect the surge and its decay – locate the 2018-late-year peak, then define the “after-effect” window as weeks where the index falls below 20 % of that peak. This follows the step-by-step GT-indicator construction in Lolić et al. (2024) 1.
5. Apply a lightweight time-series benchmark:
   * Compute Pearson or Spearman correlations between the post-surge Trend series and each auxiliary series.
   * Run a simple lagged regression (e.g., $Y_t=\beta_0+\beta_1 GT_{t-k}+\varepsilon_t$) to test whether the trend predicts outcomes k weeks later.
   * If you want a more robust test with no extra data, use the “variance-in-time” method from Mavragani et al. (2022) which exploits the intrinsic variance of the GT series to assess forecasting power 2.
6. Interpretation – a statistically significant ($p < 0.05$) coefficient or correlation that persists throughout the after-effect window indicates a lasting impact; a rapid return to zero suggests only a temporary meme.

A cyber uncertainty index based on news reports

- News reports about major incidents reflect the state of cybersecurity, and shape public opinion → might alter online behaviour.
- Economic Policy Index (Baker, Bloom and Davis, 2016)*.
- Index construction:
  1. Identify one leading newspaper by country,
  2. Define search criteria (see Table 4),
  3. Compute the relative frequency of reports talking about cybersecurity,
  4. Normalise the index.
- Index values = proportion of cybersecurity-related articles in each newspaper → relative prominence of cybersecurity in the media landscape.
- High frequency of articles on cyber-incidents → potentially high uncertainty → potentially affect investment and uptake of online services.

Table 4. Alternative search criteria for *Le Figaro*, the *FAZ* and *The New York Times*

	Search terms	English translation
Le Figaro	"cybersécurité"	"cybersecurity"
Frankfurter Allgemeine Zeitung I	"IT-Sicherheit" OR "Cybersicherheit"	"IT security" OR "cybersecurity"
Frankfurter Allgemeine Zeitung II	"IT-Sicherheit" OR "Cybersicherheit" OR "Datendiebstahl" OR "Datenleck" OR "Hackerangriff" OR "Datenpanne„	"IT security" OR "cybersecurity" OR "data theft" OR "data breach" OR "hacker attack" OR "data breach"
Frankfurter Allgemeine Zeitung III	(("IT-Sicherheit" OR "Cybersicherheit") AND ("Angriff" OR "Vorfall" OR "Panne")) OR ("Datendiebstahl" OR "Datenleck" OR "Hackerangriff" OR "Datenpanne")	(("IT security" OR "cybersecurity") AND ("attack" OR "incident" OR "breach")) OR ("data theft" OR "data breach" OR "hacker attack" OR "data breach")
The New York Times	"cybersecurity"	

* Baker, S. R., Bloom, N., & Davis, S. J. (2016), "Measuring economic policy uncertainty," *The Quarterly Journal of Economics*, 131(4), 1593-1636.

→ https://unstats.un.org/bigdata/events/2024/measuring-digital-economy/presentations/day2/6.%20LESHER_Cybersecurity_Xiamen30May2024.pdf?utm_source=chatgpt.com

If we use Googletrend data to predict trend and then compare it to actual Voting data GOOD PAPER FOR Handling GT

→ https://link.springer.com/article/10.1186/s40537-023-00868-4

→ methodology quite good for handling the Google Trends Data beforehand, although they searched for importance of keywords first :

In Step 2, we compared all search terms for a party identified in Step 1 to explore how relevant the single search terms were in relation to the other search terms. The comparison showed that the party abbreviation,

If we look at Googletrend Data and just look at the search volume and associated terms to see whether there is an increase in search or not

https://doi.org/10.1080/10807039.2017.1350566

→ Atterning changes in the search volume (synchronous increase and decrease) were detected by using the association analysis.
converted into values ranging between –1 and 1, indicating the gradual level of increase/decrease by the time.

Apriori algorithm was used to determine the associations between search volumes of the keywords. As a consequence, several patterns were detected.

If we use Google trend data and Stockmarketprice of Meta:

https://www.kaggle.com/datasets/adilshamim8/meta-stock-price-history?resource=download

→ Paper on how Data breach has an impact on stock market but rather focuses on finding connecting groups that are also affected by breach but good for looking at the impact on stock market

https://www.opastpublishers.com/open-access-articles/analyzing-the-impact-of-the-facebookcambridge-analytica-data-scandal-on-the-us-tech-stock-market-a-clusterbased-event-st.pdf

→ Paper on how search Volume can give insights on financial performance, looks at assets, the methodology quite good

https://doi.org/10.1080/10696679.2020.1867478

→ We could look whether search volume would forecast financial performance instead of looking at the longterm impact

Other papers:

https://doi.org/10.1002/for.250

Hypothesis

H_0: Weekly Google-Trends interest in the Cambridge Analytica scandal has no systematic association with weekly Facebook stock-price movements.

H_1: Higher Google-Trends interest is associated with abnormal Facebook stock-price movements (typically negative returns) during and after the surge.

Prediction (directional)

The week of the peak (late 2018) will show a statistically significant drop in Facebook returns relative to the pre-surge mean.
Weeks in the "after-effect" window (when the Trend index stays < 20 % of its peak) will still exhibit a modest but statistically significant negative correlation with returns, indicating a lasting market impact.

Analytical steps (minimal extra data)

Normalize both series to the 0-100 scale (min-max) so they are comparable.
Identify surge & decay – locate the peak of the Trend series and define the post-surge period as weeks below 20 % of that peak (method used in Lolić et al., 2024).
Lagged correlation / regression – estimate a simple lag model
*where is the weekly log-return and $\text{GT}*{t-k}$ the Trend value k weeks earlier.
Significant ($p < 0.05$) supports H_1.
Robustness – compute Spearman's ρ and Kendall's W across regions (the latter showed strong agreement ($W \approx 0.88$) in prior Google-Trends studies) to confirm consistency 1.
Dynamic-time-warping (DTW) can be added to check shape similarity between the Trend curve and the return series; DTW similarity indices of 0.60–0.79 were interpreted as meaningful alignment in earlier work 1.

If the coefficient remains negative and significant throughout the after-effect window, the public-interest surge leaves a **lasting impact** on Facebook's market performance; otherwise the effect is transient.

Methodologie Research:
https://www.jmir.org/2018/11/e270/
10.2196/jmir.9366

Keyword search Googletrends data in medical application

The aim of this review was to include all articles on the topics of health and medicine that have used Google Trends data since its establishment in 2006 through 2016. We searched for the term "Google Trends" in the Scopus [94] and PubMed [95] databases from 2006 to 2016, and following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (Figure 1), the total number of publications included in this review was 109.

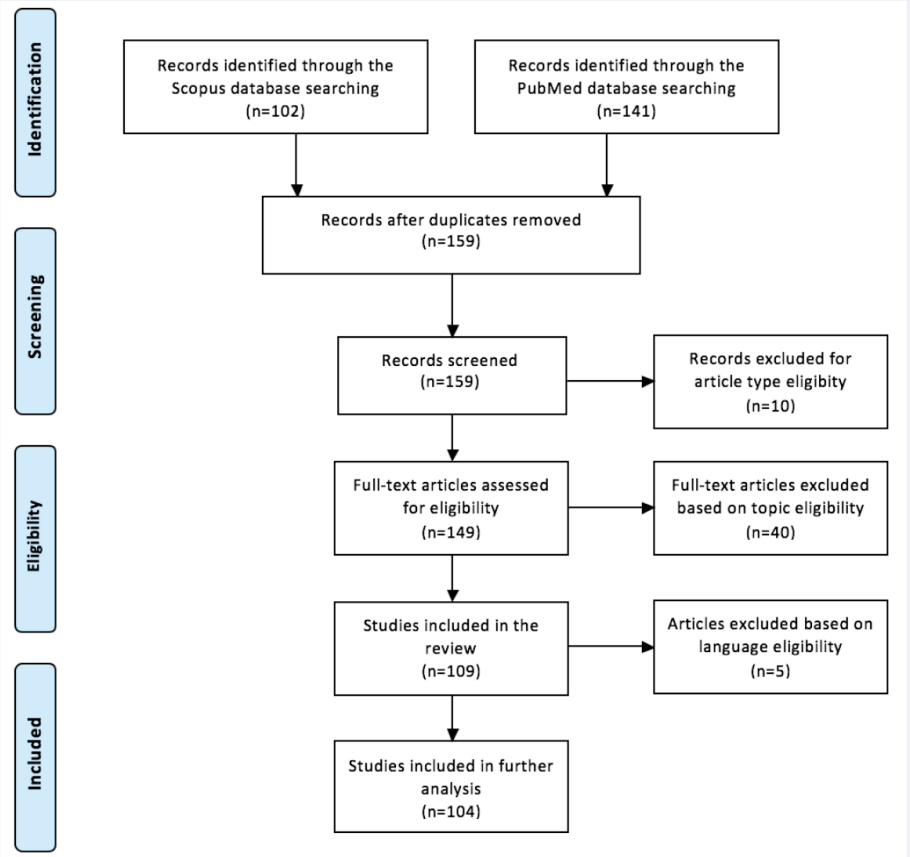

Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow diagram of the selection procedure for including studies.
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